

A Data-Centric Approach to Polyp Segmentation: Evaluating the Impact of Preprocessing and Ground-Truth Refinement

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ABSTRACT

Colorectal cancer screening plays an important role in identifying early polyps for their removal and thus requires advanced CAD algorithms. However, the challenge faced due to a variety of different shapes and imaging artifacts of polyps necessitates a data-centric perspective. Although modern Computer-Aided Diagnosis (CAD) systems employ advanced neural networks with large numbers of parameters, the importance of the data pre-processing is often overlooked. In our study, we introduce a data-centric pipeline to enhance the quality of images and ground truth labels in order to obtain competitive results by using only fundamental CNN architectures. The preprocessing step includes specular reflection compensation, Contrast Limited Adaptive Histogram Equalization (CLAHE) and gamma correction in order to improve image quality and remove imaging artifacts. Furthermore, a novel approach based on Breadth-First Search (BFS) algorithm is introduced to obtain clinically relevant three-class ground truths (Polyp, Tissue, Background). Evaluation is performed for three different neural networks (U-Net, ResU-Net, Attention U-Net). Attention U-Net model with the suggested data pipeline achieved the Dice score of 0.9525 and IoU of 0.9115 on the Kvasir-SEG dataset. Thus, our framework outperforms some high-parameter modern architectures showing that better data representation is a promising direction in medical image segmentation.

Keywords: Medical Image Analysis, Deep Learning, Image Segmentation, Mamba, Polyp Detection, Computer-Aided Diagnosis Systems

Introduction

Polyps are abnormal growths with a potential risk of developing into colorectal cancer; hence, polyp detection is the key to the prevention and treatment of colorectal cancer, which is still a leading cause of cancer-related mortality worldwide^{1,2}. Early detection ensures timely removal of polyps, thus preventing their progression towards colorectal cancer and considerably reducing mortality. However, there are numerous barriers to the detection of polyps. Polyps are often describable as being small, flat, or resembling surrounding normal tissue, making them difficult to visually detect; in addition, factors such as lack of contrast between the polyp and the tissue, combined with specular reflections from the camera during colonoscopy, complicate their detection. As a result, studies have shown that one in four polyps can be missed³. Human factors, including physical fatigue, varying experience, and different judgments further contribute to inconsistent detection rates². Hence the need to develop Computer Aided Diagnosis (CAD) systems for improving patient outcomes, reducing clinician time, and improving diagnostic accuracy.

Segmentation of tumor regions is a foundational component of these CAD systems. The polyp segmentation task involves pixel-level classification of the polyp region with respect to neighboring healthy tissue. This problem poses serious challenges due to several factors, as indicated earlier. Although state-of-the-art architectures such as the FocusU2Net⁴, Focus U-Net⁵, and multi-scale nested U-Nets with transformers⁶ have set high benchmarks, they often rely on complex designs and a high number of parameters. Recent research has demonstrated that achieving high segmentation accuracy on datasets requires not only advanced models but also attention to data quality⁷. This highlights the necessity to improve images through preprocessing methods and hence improve image quality and highlight relevant features prior to the training of deep learning models.

This study showed that even a simple U-Net model can produce a highly competitive dice score when implemented through a systematically designed processing pipeline addressing specific distortions of the image, such as specular reflections and lack of contrast. This result demonstrates an important trade-off: in many cases, a successful preprocessing approach can have as much effect on obtaining improved segmentation accuracy as architectural sophistication. Therefore by shifting the focus from an architectural to a data-centric approach, the study highlights the necessity to improve data through preprocessing techniques and refining the ground truth to more accurately represent the data in hand. The main contributions of this study are:

- This study introduces a structured image pre-processing and enhancement framework specifically created for colonoscopy. By systematically incorporating specular reflection compensation, CLAHE and gamma correction.
- This study identified a critical limitation in traditional binary polyp masks, which incorrectly merged biological tissue and dark, non-anatomical frame borders into a single class. To rectify this, a BFS based refinement algorithm was introduced that isolates non-anatomical background regions.
- This study performs a comprehensive evaluation on all three basic models of U-net. The proposed technique facilitates achieving Dice Score of 0.9525 and IoU of 0.9115.
- This study presents a detailed analysis of the precision vs recall trade-off made possible by the proposed post-processing techniques. While overlap metrics face slight decline, this study demonstrated that it considerably helps reducing false positives heading to higher PCP up to 0.9303

The remainder of this paper is organised as follows: Section 1 reviews the related work. Section 2 describes the proposed methodology. Section 3 presents the experimental results, and section 4 discusses the findings obtained from experiments conducted on public dataset. Finally, Section 5 concludes the paper and outlines potential directions for future research.

1 Literature Review

The U-Net Foundation and Variants

The U-Net architecture is considered the gold standard for polyp segmentation due to its symmetric encoder-decoder design and skip connections, which combine high-level semantic information with low-level spatial details^{8,9}. This architecture preserves fine features crucial for delineating polyp boundaries. The present study evaluates three primary variants: the vanilla **U-Net**¹⁰, **ResU-Net**¹¹, which utilizes residual blocks to mitigate vanishing gradients, and **Attention U-Net**¹², which incorporates attention gates. Other advanced iterations like UNet++¹³ and U-Net v2¹⁴ have further refined these connections to reduce semantic gaps, improving performance on varying object sizes.

Limitations and Long-Range Dependencies

Standard Convolutional Neural Networks (CNNs) have inherent limitations in capturing long-range dependencies due to their restricted receptive field size. This locality restricts the model's ability to integrate distant but relevant features, which is critical when polyps vary in size or lack contrast with surrounding tissue⁶. To overcome this, hybrid models combining CNNs with transformers or attention mechanisms have been proposed to explicitly model global relationships while preserving local feature extraction^{15,16}.

Attention Mechanisms and Hybrid Architectures

Attention Gates help by suppressing irrelevant regions and enhancing salient features related to the polyp, effectively filtering out background noise^{5,17}. Furthermore, hybrid architectures like SwinE-Net leverage CNNs for local details and Vision Transformers (ViTs) for global context¹⁸. However, these models demand substantial computational resources due to their high parameter complexity, limiting real-time deployment on edge devices^{19,20}.

Imaging Artifacts and Data Scarcity

Common artifacts like specular reflections, motion blur, and low contrast are primary drivers of False Negatives in AI models^{21,22}. Historically, data scarcity has been addressed through augmentation and self-supervised learning (SSL)²³. However, these strategies often fail when ground truth labels are noisy or inaccurate. This study shows that investing in data quality and label refinement—specifically moving toward three-class ground truth—can be more impactful than increasing model complexity alone^{24,25}.

2 Methodology

Dataset Refinement

The experiments were performed on the Kvasir-SEG dataset, consisting of 1000 high-resolution colonoscopy images and their ground-truth masks²⁶. There exists a fundamental limitation in its standard 2-class (polyp or non-polyp) mask representation. This method tends to classify all non-polyp areas as one class, including both the adjacent biological tissue and the dark, non-anatomical periphery of the images. From the clinical viewpoint, these black areas are not tissue. Categorizing them as such is medically incorrect and can result in erroneous outcomes for CAD systems. Hence, it was hypothesized that a stronger segmentation model should distinguish between these different classes. Towards this end, the dataset was re-labeled to 3-class

type, including Polyp, Tissue, and Black Background. To perform labeling, this study employed a Breadth-First Search (BFS) algorithm to detect and rightly reclassify these black areas in the border of the images, so that the new ground-truth masks better reflect actual anatomical tissue. The logic for this refinement is detailed in Algorithm 1.

Algorithm 1 BFS-based Ground Truth Refinement

```

1: Input: Original Image  $I$ , Original Mask  $M$ 
2: Output: Refined 3-Class Mask  $M_3$ 
3: Initialize  $M_3 \leftarrow M$ ,  $Visited \leftarrow$  False matrix
4:  $Queue \leftarrow$  empty queue
5:  $T_{black} \leftarrow 15$  {Threshold for dark pixels}
6:  $Corners \leftarrow \{(0,0), (0, w-1), (h-1,0), (h-1, w-1)\}$ 
7: for each  $(y,x)$  in  $Corners$  do
8:   if  $\max(I_{y,x}) \leq T_{black}$  AND  $M_{y,x} == 0$  then
9:      $Queue.push(y,x)$ ,  $Visited_{y,x} \leftarrow$  True
10:     $M_3(y,x) \leftarrow Class_{Background}$ 
11:   end if
12: end for
13: while  $Queue$  is not empty do
14:    $(cy,cx) \leftarrow Queue.pop()$ 
15:   for each neighbor  $(ny,nx)$  using 8-connectivity do
16:     if  $(ny,nx)$  in bounds AND NOT  $Visited_{ny,nx}$  then
17:       if  $\max(I_{ny,nx}) \leq T_{black}$  AND  $M_{ny,nx} == 0$  then
18:          $Visited_{ny,nx} \leftarrow$  True
19:          $M_3(ny,nx) \leftarrow Class_{Background}$ 
20:          $Queue.push(ny,nx)$ 
21:       end if
22:       if NOT  $Visited_{ny,nx}$  AND  $M_{ny,nx} == 0$  then
23:          $M_3(ny,nx) \leftarrow Class_{Tissue}$ 
24:       end if
25:     end if
26:   end for
27: end while
28: return  $M_3$ 

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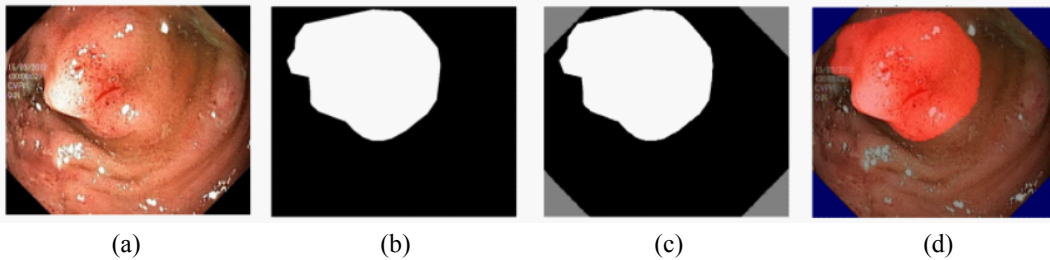


Figure 1. Comparison of 2-class and 3-class Ground Truth Masks. **(a)** Original Colonoscopy Image. **(b)** Original 2-Class Ground Truth, where the polyp area is white and all non-polyp regions are black. **(c)** Suggested 3-Class Ground Truth, identifying three classes: polyp (white), tissue (black), and non-anatomical background (gray). **(d)** 3-Class Mask Overlay with lower opacity upon the original image; the polyp is highlighted in red and the non-anatomical background in blue.

Fig. 1 shows the essential difference between the traditional 2-class ground truth and the improved 3-class segmentation labels. Left to right: (a) Original Colonoscopy Image: A sample image from the Kvasir-SEG dataset, with a polyp. (b) Original 2-Class Ground Truth: The ground truth mask given by the dataset, in which the polyp area is white and all non-polyp regions, including black background and surrounding tissue, are black. (c) Suggested 3-Class Ground Truth: The proposed ground truth mask, identifying three classes: the polyp (white), the tissue (black), and the non-anatomical background area (gray).

(d) 3-Class Mask Overlay: The three-class mask is overlaid with lower opacity upon the original image to enable meaningful visualization. The polyp is highlighted in red, while the non-anatomical background is highlighted in blue.

Image Processing

The low-contrast, nonuniform lighting, and specularity reflections in a colonoscopy image were addressed by a multi-step preprocessing pipeline. This pipeline enhances the raw image quality and contributes to the robustness of the trained models. The employed sequence consists of three phases, each of which was designed to mitigate specific image artifacts.

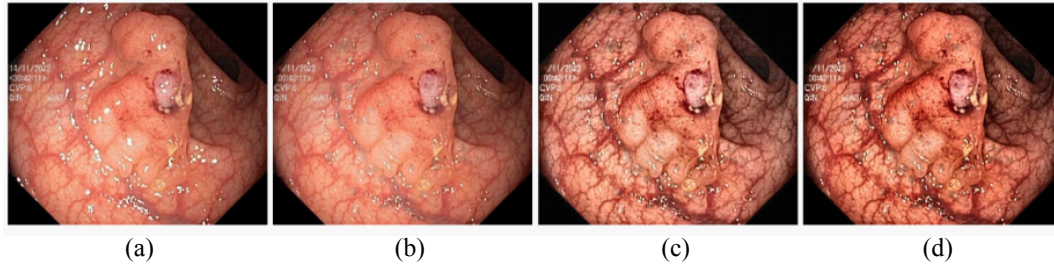


Figure 2. Multistage Image Preprocessing Pipeline. **(a)** Original Image, showing significant glare and low contrast. **(b)** Specular Reflection Correction, where bright artifacts are successfully removed. **(c)** CLAHE Application, showing improved local contrast and visibility of subtle features. **(d)** Gamma Correction, where the overall brightness and intensity are adjusted for a more uniform appearance.

Fig. 2 illustrates the effect of these below-mentioned steps on the image. From left to right: (a) Original Image, showing significant glare and low contrast; (b) Specular Reflection Correction, where bright artifacts are successfully removed²⁷; (c) Contrast Limited Adaptive Histogram Equalization (CLAHE) Application, showing improved local contrast and visibility of subtle features²⁸; and (d) Gamma Correction, where the overall brightness and intensity are adjusted for a more uniform appearance.

Model Architecture

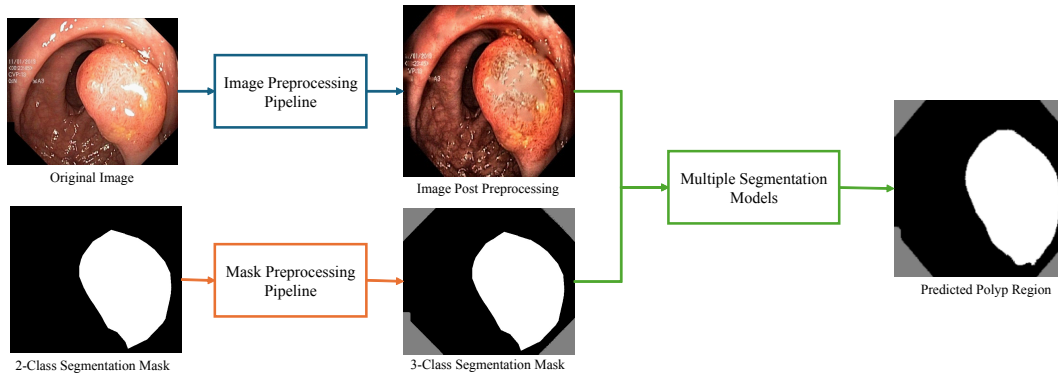


Figure 3. Polyp Segmentation Process Overview.

An overview of the polyp segmentation is described in Fig. 3. The segmentation models in this study are based on the U-Net architecture¹⁰, which is a widely adopted deep learning framework for biomedical image segmentation due to its effective encoder-decoder structure and skip connections. The specific U-Net architecture used for this study is shown in Fig. 4.

This study specifically evaluated three key variants: the foundational U-Net, the ResU-Net which incorporates residual connections (as shown in Fig. 5) to enhance feature propagation and combat vanishing gradients²⁹, and the Attention U-Net that uses attention gates (as shown in Fig. 6) to selectively focus on relevant features, improving segmentation accuracy in complex cases¹². These models were chosen to systematically assess the impact of key architectural enhancements on performance within our proposed data pipeline.

To fairly evaluate all the models, the main architecture with each model remains the same with minor changes in each block, Fig. 5 demonstrates the Encoder and Decoder blocks that were implemented in the Res U-Net model trained for this study.

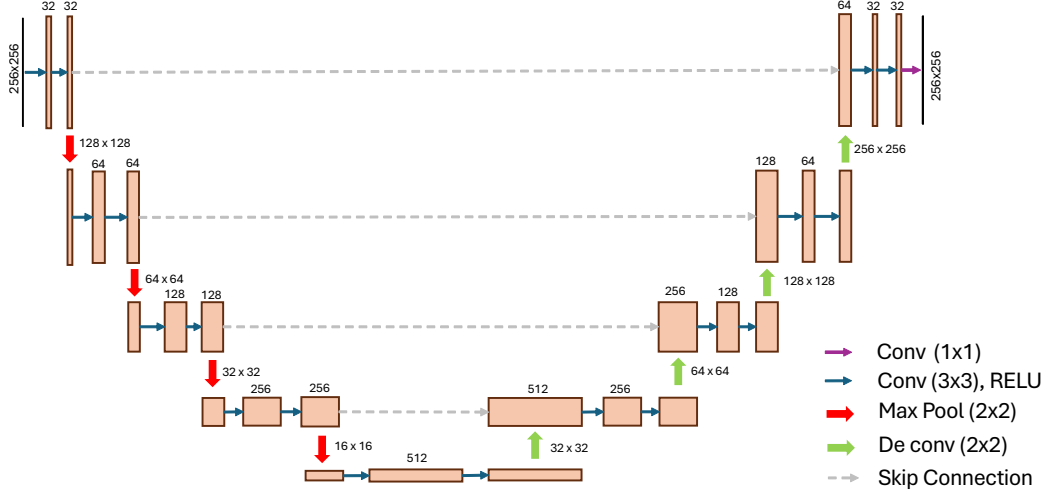


Figure 4. Baseline U-Net architecture used in this study.

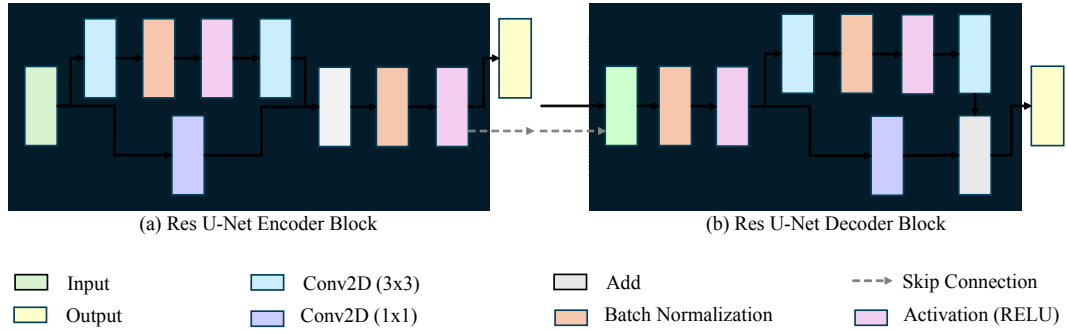


Figure 5. Encoder and Decoder blocks implemented in Res U-Net.

The Attention U-Net architecture incorporates attention gates in between the skip connection and gating signals. Fig. 6 represents two inputs, being Input 1 and Input 2 respectively, where Input 1 is the information from the Skip Connection ‘x’ while Input 2 is the Gating Signal ‘g’ from the Conv2DTranspose (up-sampled) layer.

Training Environment

All experiments were conducted on a machine with a NVIDIA RTX 3050 Laptop GPU and a Windows 11 operating system. The models were implemented using Python 3.9 and the TensorFlow 2.10 framework. For GPU acceleration, CUDA Toolkit 11.2 and cuDNN 8.1.0 were utilized. The core libraries used for data handling and processing were NumPy 1.26 and OpenCV 4.11.0.86.

Training Parameters

The training of all three models was performed using a consistent set of parameters. Each model was run for a maximum of 100 epochs, with a batch size of 2. All input images were resized to 256×256 pixels with three color channels. Adam optimizer was used for optimization with an initial learning rate of 1×10^{-4} . Early stopping with a patience of 15 epochs and Model Checkpoint callbacks were implemented. Learning rate was reduced by a factor of 0.5 if the validation loss of the model did not improve after 7 epochs.

Sparse Categorical Crossentropy was used as the loss function, as it is widely used for multi class segmentation with integer-encoded labels. The loss was calculated using equation (1):

$$Loss = - \sum_{i=0}^{C-1} y_i \log \hat{y}_i \quad (1)$$

Where C is the number of unique classes, y_i is the binary indicator for class i and $\hat{y}_i$ is the predicted probability of class i .

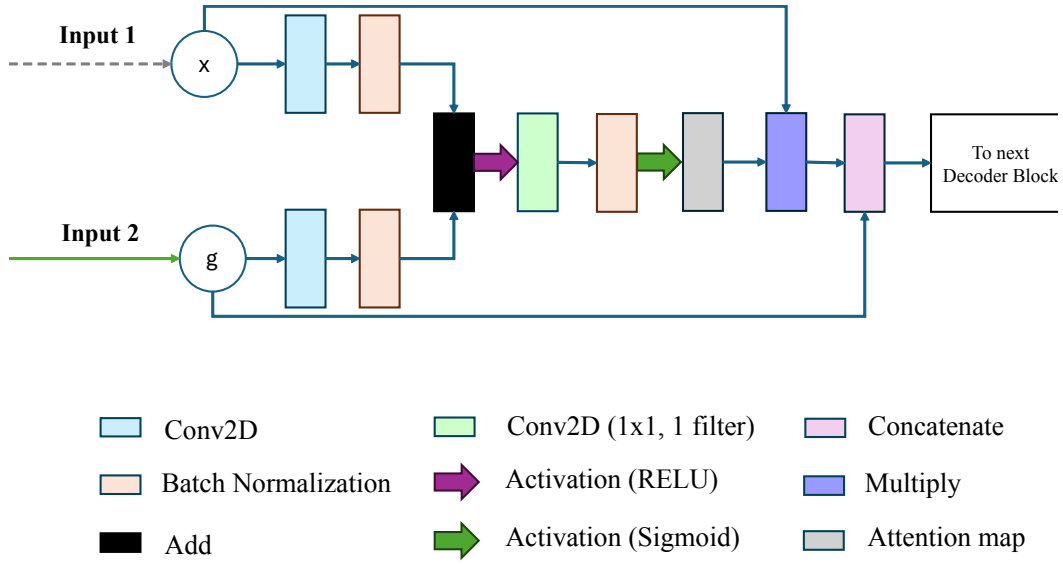


Figure 6. Attention Gate incorporated in this study.

Evaluation Metrics

The performance of the models was assessed using the Dice Coefficient³⁰ and Jaccard Index³¹ Metrics.

Dice Coefficient. The measure of similarity between predicted and ground truth masks³⁰. The formula for a given class k is:

$$Dice_k = \frac{2 \times \sum_{i=1}^N p_{ik} y_{ik}}{\sum_{i=1}^N p_{ik} + \sum_{i=1}^N y_{ik}} \quad (2)$$

Intersection over Union (IoU). The Jaccard index (IoU) quantifies the overlap between the predicted and true masks³¹. The formula for a given class k is:

$$IoU_k = \frac{\sum_{i=1}^N p_{ik} y_{ik}}{\sum_{i=1}^N p_{ik} + \sum_{i=1}^N y_{ik} - \sum_{i=1}^N p_{ik} y_{ik}} \quad (3)$$

3 Results

Quantitative Performance Analysis

Performance of the three variants of U-Net was extensively tested under nine conditions of training, in which the three models were trained on raw and preprocessed images. Furthermore, for the models that were trained with a pre-processing pipeline, to evaluate the quantitative impact of the post-processing step, two more evaluations were performed, one where post-processing was applied, and one without post-processing.

Overall performance measures under all the experimental conditions are presented in Table 1. The table indicates consistent improvement in performance because of preprocessing and the additional advantage because of post-processing methods. The ‘PP’ column signifies application of the preprocessing pipeline, and ‘PSP’ signifies application of the post-processing on the model output. The performance evaluation metrics employed are the Dice Coefficient and Intersection over Union (IoU) to measure overall segmentation accuracy, and PCP (Polyp Class Precision) and PCR (Polyp Class Recall) for the polyp class in the multi-class segmentation model.

Qualitative Results

Fig. 7 and Fig. 8 provide a detailed visual analysis of the proposed segmentation approach. Fig. 7 illustrates the end-to-end impact of the preprocessing and post-processing pipelines on the final segmentation output. This high-level comparison is further supported by Fig. 8, which provides a more granular examination of specific challenging regions, highlighting how each step of the pipeline addresses common image artifacts to improve model performance and accuracy.

Fig. 8, a multi-panel figure provides an in-depth visual comparison of model behavior and segmentation quality.

Table 1. Comparative Performance of U-Net Variants with and without Preprocessing and Post-Processing. PP: preprocessing pipeline applied; PSP: post-processing applied; PCP: Polyp Class Precision; PCR: Polyp Class Recall.

Model	PP	PSP	Dice	IoU	PCP	PCR
U-Net	No	No	0.8833	0.7959	0.8663	0.7603
U-Net	Yes	No	0.9286	0.8706	0.7867	0.8481
U-Net	Yes	Yes	0.9232	0.8611	0.9303	0.6220
ResU-Net	No	No	0.8851	0.7993	0.7683	0.5542
ResU-Net	Yes	No	0.9430	0.8957	0.8769	0.8063
ResU-Net	Yes	Yes	0.9384	0.8877	0.9347	0.7190
Attention U-Net	No	No	0.9299	0.8735	0.8724	0.7558
Attention U-Net	Yes	No	0.9525	0.9115	0.8921	0.8545
Attention U-Net	Yes	Yes	0.9514	0.9098	0.9253	0.8143

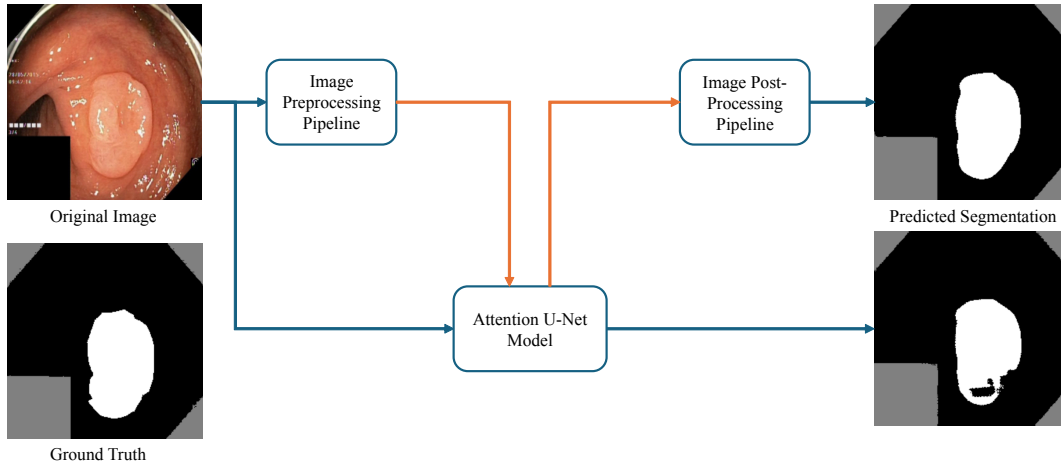


Figure 7. Illustration of the End-to-End Segmentation Pipelines. The top path shows the flow for the preprocessed and post-processed approach, where an original image is first enhanced before being segmented. The bottom path shows the results from a model trained and tested without these additional steps. The clear difference in the segmented outputs highlights the cumulative performance gain of the proposed pipeline.

Part (a): This panel shows the final prediction of the preprocessed and post-processed model overlaid on the original image. The accompanying Grad-CAM heatmap from the final layer demonstrates a strong, precise focus on the polyp, indicating effective feature learning.

Part (b): This panel displays the prediction from the model without preprocessing or post-processing overlaid on the original image. The corresponding Grad-CAM heatmap reveals a less focused or more scattered attention pattern, which correlates with the less accurate segmentation.

Part (c): This close-up view focuses on a specular reflection region. A comparative view of the Grad-CAM heatmaps shows the model trained on preprocessed data has no attention on the bright specular reflection. In contrast, the model trained on raw data shows a small amount of focus on this non-anatomical feature, demonstrating the importance of specular reflection correction in training.

Part (d): This panel highlights a difficult-to-segment region of the polyp boundary. The Grad-CAM heatmap for the non-preprocessed model shows a weak and scattered focus, corresponding to a missed prediction. The Grad-CAM for the preprocessed model shows a strong and concentrated focus in the same area, which successfully led to the correct segmentation. This visually demonstrates the critical role of contrast enhancement and gamma correction in defining challenging boundaries for the model.

Figure 9 demonstrates the effectiveness of the post-processing pipeline employed; this provides a qualitative analysis of the model's confidence maps. This approach visually explains how low-confidence, spurious predictions are filtered out to produce a cleaner, more accurate final mask.

Figure 9 provides a visual breakdown of the post-processing pipeline using confidence maps from the Attention U-Net

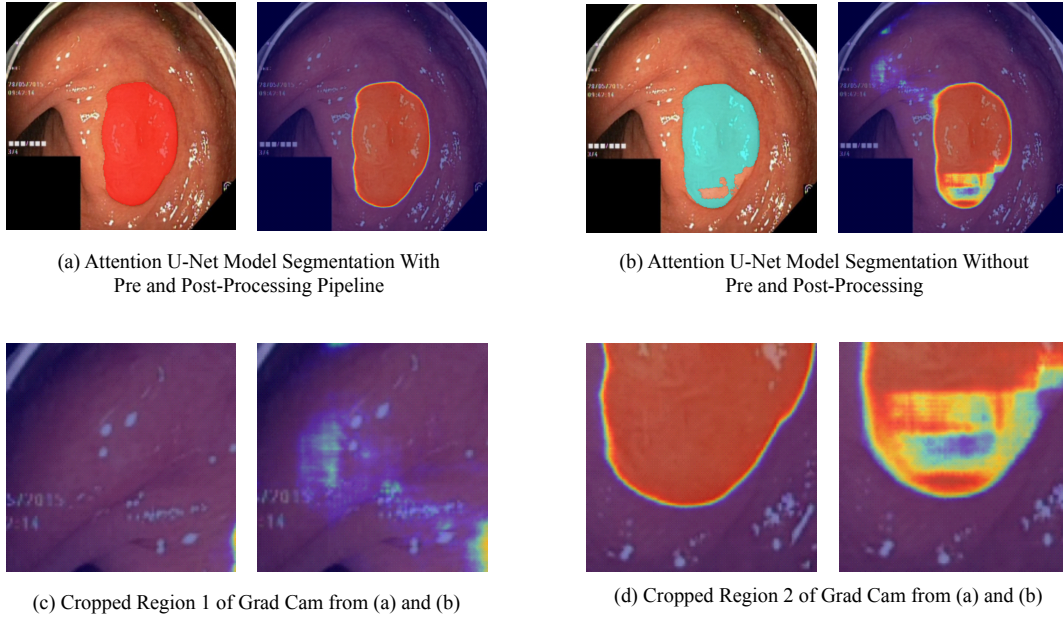


Figure 8. Detailed Qualitative and Experimental Analysis. **(a)** Final prediction of the preprocessed and post-processed model with Grad-CAM heatmap showing precise polyp focus. **(b)** Prediction without preprocessing showing scattered attention. **(c)** Close-up of specular reflection region comparing Grad-CAM attention between preprocessed and raw models. **(d)** Challenging polyp boundary region demonstrating improved focus with preprocessing.

model. Part (a) shows the preprocessed original image. Part (b) displays the confidence map of the model's raw prediction, where pixel values represent the model's confidence in its classification. Note the presence of a small purple patch, representing low-confidence predictions, in a region that is not a polyp. Part (c) shows the confidence map of the final prediction after applying our post-processing pipeline. The absence of the low-confidence purple patch demonstrates how the pipeline successfully filters out spurious predictions, resulting in a more precise and clinically reliable segmentation.

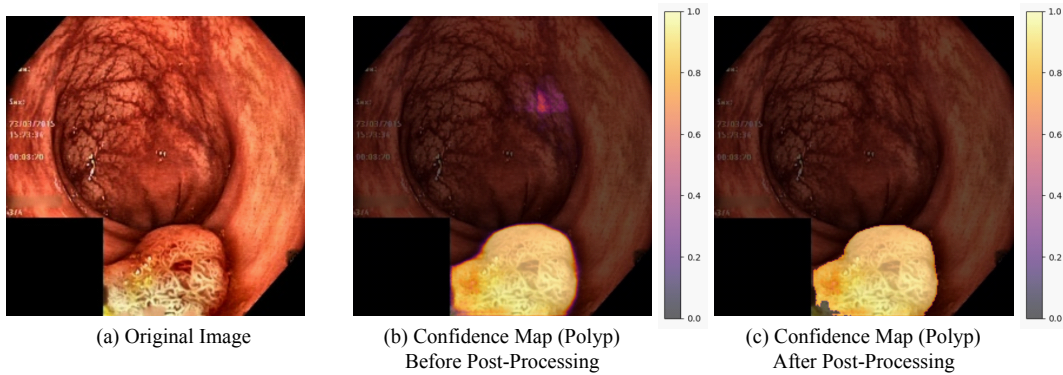


Figure 9. Analysis of Confidence Maps Demonstrating the Impact of Post-Processing. **(a)** Preprocessed original image. **(b)** Confidence map of raw model prediction; the small purple patch indicates low-confidence spurious predictions in a non-polyp region. **(c)** Confidence map after post-processing; the spurious patch is eliminated, yielding a more precise and clinically reliable segmentation.

Comparison with State-of-the-Art Architectures

The enhanced performance of the suggested approach is revealed when contrasted with the current state-of-the-art (SOTA) approaches reported in the literature. The comparison is important to validate the approach and emphasize its contribution. Although the majority of the SOTA approaches address the classical 2-class segmentation task with the classical Kvasir-SEG masks, the suggested approach tackles a more difficult 3-class task with a more informative and clinically useful output.

ResUNet++³²: This network, being the direct inspiration behind most modern architectures, was trained on the standard 2-class Kvasir-SEG dataset. It achieved a reported Dice coefficient of 0.8133 and an mIoU of 0.7927. This performance is substantially surpassed by the models presented in this study.

HarDNet-based Model³³: In this model, regular 2-class segmentation was applied to the Kvasir dataset, and it achieved a Dice of 0.915 and an IoU of 0.862. These are strong results but remain lower than the performance achieved by the present method.

ICSO-AFConvNet³⁴: On the Kvasir-SEG 2-class dataset, the model has a Dice of 0.941 and an IoU of 0.906. This is a highly competitive result, but marginally below that of the top-performing model in this study.

PolySeg Plus³⁵: This was also evaluated on a 2-class segmentation task on Kvasir-SEG. It achieved a Dice of 0.9476, which showed strong performance.

PNS+³⁶: This model is specifically designed for complex task of video polyp segmentation (VPS). The model had a Dice coefficient of 0.756 on the SUN-SEG dataset, and direct comparison is impossible, but the models of this study are still in a good position.

Table 2. Performance Comparison of the Proposed Model Against State-of-the-Art Approaches. All SOTA models evaluated on the standard 2-class Kvasir-SEG task; our model addresses the harder 3-class task.

Model	Dice	IoU
Our Model (3-Class)	0.9525	0.9115
PolySeg Plus (2-Class)	0.9476	-
ICSO-AFConvNet (2-Class)	0.941	0.906
HarDNet-based (2-Class)	0.915	0.862
ResUNet++ (2-Class)	0.8133	0.7927
PNS+ (Video Seg.)	0.756	-

Table 2 gives a clear overview, illustrating how the Attention U-Net model with the suggested preprocessing and post-processing pipeline performs better. The results prove that even when using a multi-class segmentation method, the model performs well, highlighting the clinical benefit of separating relevant tissue from non-anatomical structures like camera obstructions or a narrow field of view.

4 Discussion

The experimental results demonstrate that the proposed pipeline, which integrates preprocessing and post-processing with U-Net variants, is highly effective for polyp segmentation. The quantitative analysis confirmed a consistent performance gain when models were trained on preprocessed data, a finding supported by the qualitative analysis (Fig. 8) where Grad-CAM visualizations showed models focusing more precisely on the polyp rather than on image artifacts.

One of the observations from the quantitative analysis is the precision-recall trade-off due to the post-processing pipeline. Although the Dice Coefficient and IoU metrics, being a harmonic mean of the two, saw a marginal drop due to post-processing, a closer look into the polyp class metrics (PCP and PCR) shows a deliberate and visible shift. As shown in Table 1, post-processing, for all the models, shows enhanced precision (PCP) and reduced recall (PCR). For instance, the PCP for the U-Net model increased from 0.7867 to 0.9303, showing a drastic reduction in false positives. In the context of a clinical scenario, such a trade-off can be highly desirable, where the confidence level of the segmentation comes at the expense of the polyp regions identified being highly credible and reducing the likelihood of false alarms leading to unnecessary follow-up interventions.

The comparative performance on the three architectures again demonstrates the superiority of the Attention U-Net. It consistently generated the highest Dice and IoU scores, both with and without post-processing. Its performance can be attributed to the attention mechanisms inherent in it, enabling it to learn more discriminative features and make more consistent, high-confidence predictions. Its drop in Dice score after post-processing by a small extent indicates that its raw output was already better than the others, needing less aggressive filtering. The ResU-Net was also well-performing, reaffirming that residual connections are very helpful for medical image segmentation.

Despite these results, there are some limitations and future work avenues. The models were validated on one, albeit widely used, dataset. Future work can investigate how the pipeline generalizes to other colonoscopy datasets with other camera sensors and image features. Further, exploring the application of this work to real-time video polyp segmentation is a vital next step, tackling the analysis challenge in dynamic clinical settings.

5 Conclusion

This study presents a holistic polyp segmentation framework that combines a custom preprocessing and post-processing pipeline with well-known variants of U-Net. The proposed methodology addresses salient colonoscopy image challenges, common artifacts and refining model predictions. Experimental results demonstrate significant enhancements in the performance of U-Net, ResU-Net, and Attention U-Net under this framework. In particular, an Attention U-Net model achieved a Dice score of 0.9525 and an IoU of 0.9115, setting a new state-of-the-art benchmark for this segmentation task. A core contribution of this work is the clinically more relevant three-class segmentation, which our models successfully performed while outperforming existing state-of-the-art models, which were trained on a simpler two-class binary task. Our results highlight that an optimized data pipeline is often more impactful than pure architectural complexity and allow for further development toward more accurate, efficient, and clinically relevant deep learning models in medical image analysis.

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Author contributions statement

Akshat G contributed to the implementation of the proposed algorithms, experimentation, and data analysis. Dr. Rashmi R conceptualized the study, supervised the research, and contributed to the methodology development and manuscript preparation. Both authors interpreted the results, reviewed the manuscript critically, and approved the final version for publication.

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Declarations

Competing Interests The authors declare no competing interests.

Additional information

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Data Availability. The Kvasir-SEG dataset used in this study is publicly available at <https://datasets.simula.no/kvasir-seg/>.